Chapter 5: Errors, Power, and Effect Size

Statistics for Environmental Science

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> *It is better to be roughly right than precisely wrong.*
>
> Attributed to John Maynard Keynes

Chapter 4 ended with a hypothesis test that either rejects H_0 or fails to reject it.

This chapter first asks how often that decision is wrong, and in which direction. It then asks how large the effect is and whether the study had enough data to detect it. It closes with how to report a result.

Errors in hypothesis testing

Sometimes, even when we apply the test correctly, sampling variability can lead us to the wrong conclusion.

- **Type I error:** rejecting H_0 when H_0 is true. It is a false positive, or false alarm, like a smoke alarm going off when there is no fire. If H_0 is true, the probability of a Type I error is α, the significance level from Chapter 4. By convention, α is often set at 0.05 (5%).
- **Type II error:** failing to reject H_0 when H_0 is false. It is a missed signal, like a smoke alarm staying silent when there *is* a fire. The probability of a Type II error is β. Its value depends on how far the true mean is from μ_0, which the figures below show.

An error here means the decision does not match the truth, at a rate the design sets. It is not a mistake by the person running the test.

The table crosses the two possible decisions with the two possible truths, giving two correct outcomes and two errors.

Decision	If the null hypothesis is True	If the null hypothesis is False
Reject H_0	Type I error (prob $= \alpha$)	Correct (prob $= 1 - \beta$, the power)
Fail to reject H_0	Correct (prob $= 1 - \alpha$)	Type II error (prob $= \beta$)

The probability $1 - \beta$ of correctly rejecting a false H_0 is the power of the test (defined in the Working definitions later in this chapter).

For a fixed sample size and effect size, lowering α raises β. A smaller α moves the critical values farther from μ_0, so more samples from a population with a real difference fall short of them, and the test fails to reject. Rows 2 and 3 of Table 3 later show this in numbers for one design.

Which error costs more depends on the study. Table 1 gives the cost of each error for three tests. The first two appear later in this chapter.

Table 1: The cost of each error in three tests.

Test	Type I error here	Type II error here	Which is worse here (a judgment call)
No-idling rule near schools, H_0: the rule did not change PM2.5	Keep a rule that does nothing	Drop a rule that works	Here, Type II would be worse. Kids would breathe idling exhaust that the rule would have cut.
Discharge permit limit, H_0: $\mu = 10$ mg/L	Act on an exceedance that is not there	Keep operating while the discharge exceeds the limit	Here, Type II is worse for the river, which keeps getting the excess discharge. Type I is probably worse for the plant, which pays to fix a problem it does not have.

Test	Type I error here	Type II error here	Which is worse here (a judgment call)
Removing a species' protected status, H_0: the population has not recovered	Remove protection from a population that has not recovered	Keep protecting a population that has recovered	Here, Type I would be worse. Losing protection too early could send the population back into decline, while extra years of protection mostly cost money and paperwork.

Figure 1 recaps the test from Chapter 4. It shows two sampling distributions of the sample mean. The blue one is centered on μ_0 and shows the sample means we would expect if H_0 were true. The green one is centered on our sample's $\bar{x}$. The distance between their centers, in standard errors, decides whether we reject H_0. As in Chapter 4, the figures treat σ as known, so their standard error is $\sigma/\sqrt{n}$. The simulations later in the chapter run t-tests, which estimate it with $s/\sqrt{n}$ (Chapter 6).

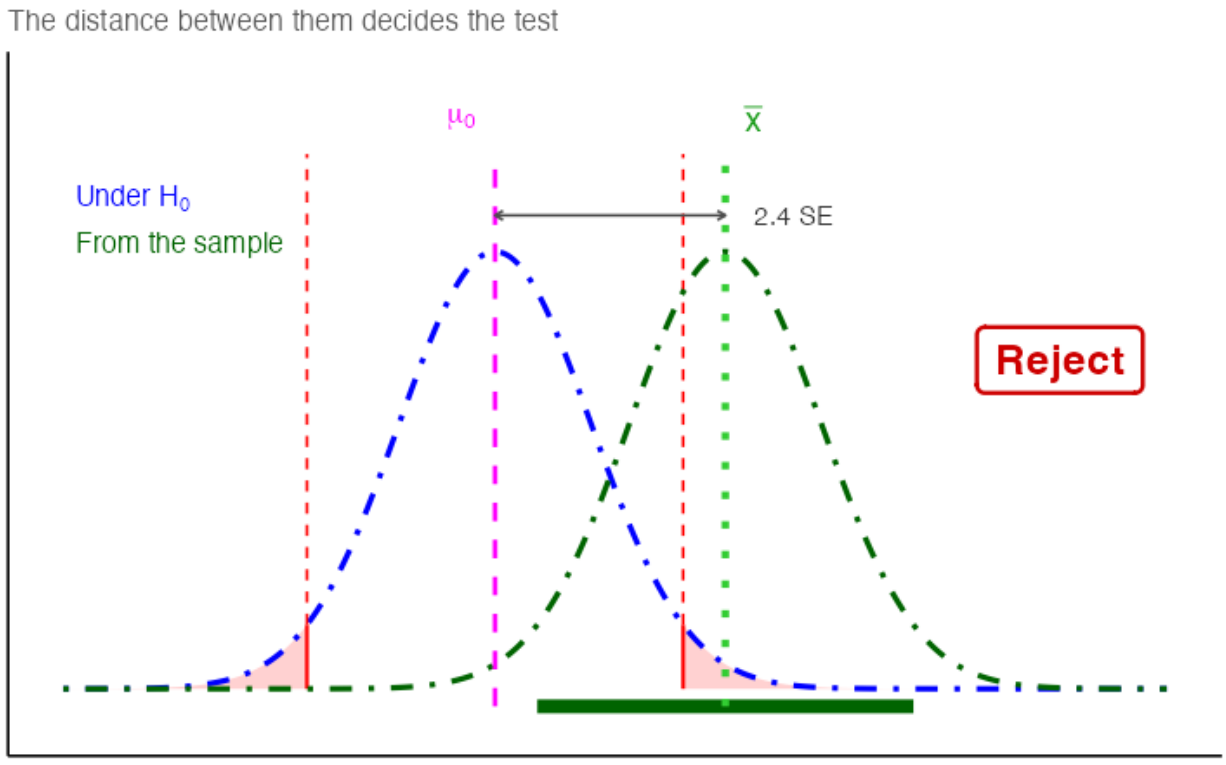


Figure 1: The two sampling distributions of the sample mean from Chapter 4. The distance between them, in standard errors, decides the test. When the distance passes 1.96, the test rejects H_0. (Final frame. The animated version is in the web edition of this book.)

- In Figure 2, the true population mean (orange) is the same as μ_0. By chance, however, the sample mean is 2.4 standard errors from μ_0, past the critical value at 1.96 standard errors, and the 95% confidence interval (CI, Chapter 4) excludes μ_0. The test rejects H_0 even though it is true, which is a Type I error.
- In Figure 3, the true population mean (orange) is 4 standard errors from μ_0. By chance, however, the sample mean is only 1.5 standard errors from μ_0, inside the critical value, and the CI includes μ_0.

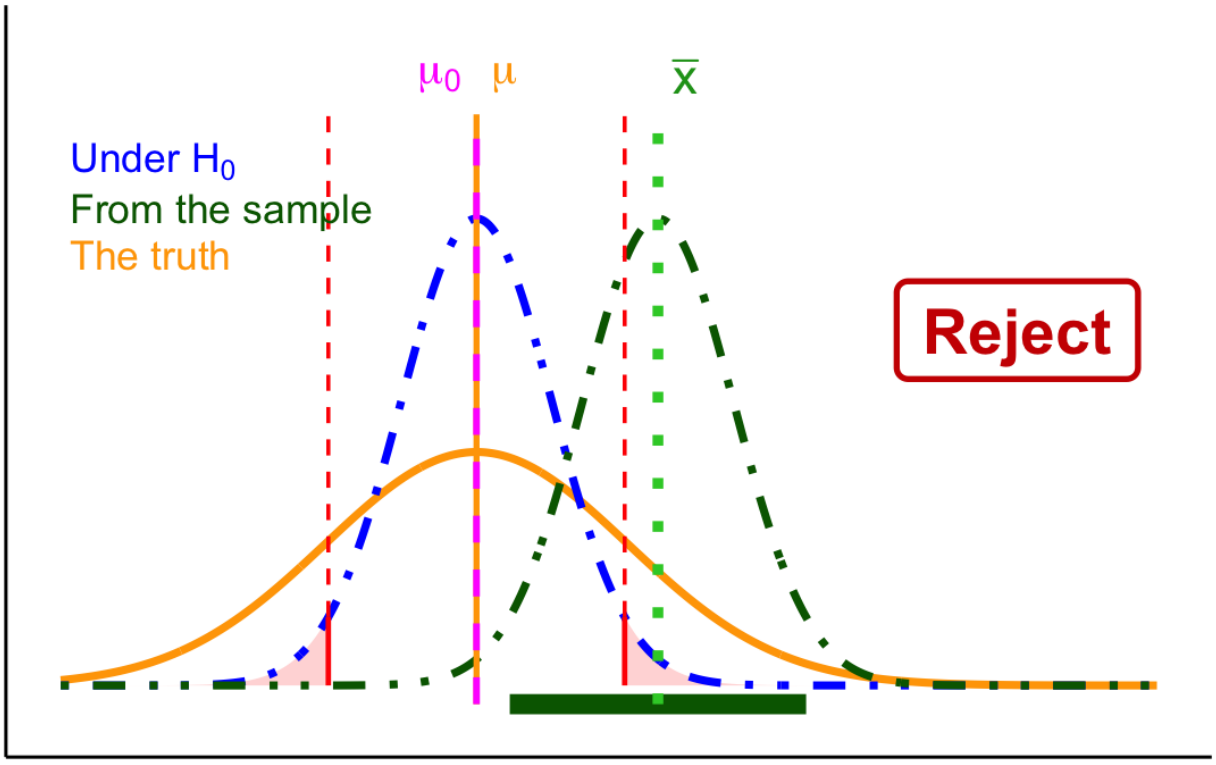


Figure 2: Type I error: H_0 is true ($\mu = \mu_0$), but the sample leads us to reject H_0.

The test fails to reject H_0 even though it is false, which is a Type II error.

These two bullets describe each decision two ways. For a two-tailed test at $\alpha = 0.05$, the sample mean falls past a critical value exactly when the 95% confidence interval excludes μ_0. The critical values are drawn around μ_0 and ask whether $\bar{x}$ is far from it. The confidence interval is drawn around $\bar{x}$ and asks whether μ_0 is far from it. Both use the same distance, 1.96 standard errors.

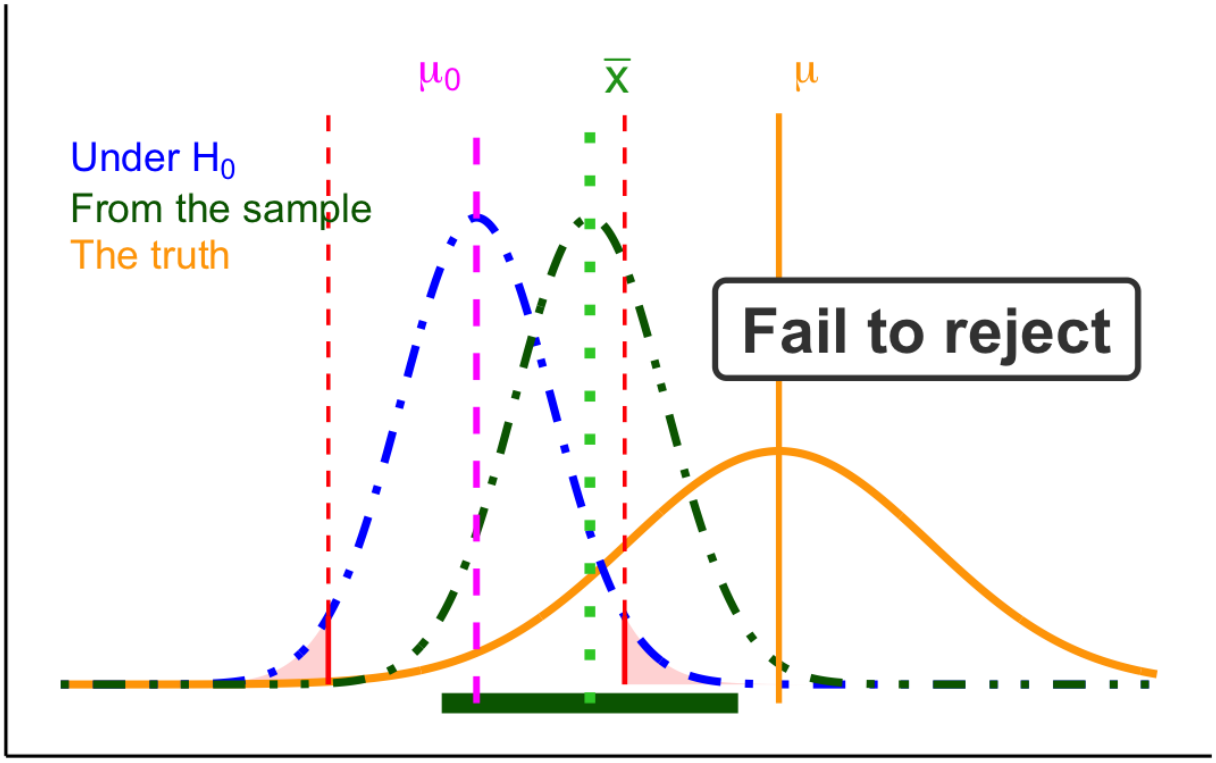


Figure 3: Type II error: H_0 is false (true population mean differs from μ_0), but the sample leads us to fail to reject H_0.

Those two figures show what a wrong conclusion looks like for one unlucky sample. They do not show how *often* it happens. The green curve in them is built from that one sample, centered on its $\bar{x}$. To see how often an error happens, center the sampling distribution on a possible value of the true mean instead. That curve shows how far from μ_0 the sample mean would be across all the samples we could have drawn if that value were the true mean.

Chapter 4 drew the possible true populations under H_a as a family of curves on either side of μ_0, and μ_a is the center of one of them. In Chapter 4's terms, the curve centered on μ_a is one possible true population distribution. The estimated population distribution is a different curve, built from the sample. μ_a is a value we choose, usually the smallest change that would alter what we do. Because we never know the true μ, we ask how the test would perform if μ_a were true. μ_a is not the sample mean or a critical value. The farther μ_a is from μ_0, the less likely the test is to miss the difference.

Figure 4 centers the sampling distribution on μ_a. The blue curve is the sampling distribution of the sample mean under H_0, the one that sets the critical values. The orange curve is the sampling distribution of the sample mean if $\mu = \mu_a$. Because it shows sample means, it has the same width as the blue curve, the standard error $\sigma/\sqrt{n}$. These figures use $n = 4$, so the standard error is $\sigma/\sqrt{4} = \sigma/2$, and the orange curve here is half as wide as the orange population curve in the last two figures.

- α is the red area in the tails of the blue curve, past the critical values. If H_0 is true, α is how often the sample mean falls in the rejection region by chance. Computing α needs only the blue curve, so the top panel has no orange curve. The orange curve appears in the bottom panel, for β.
- β is the gray area under the orange curve *between* the critical values. If $\mu = \mu_a$, that is how often the sample mean fails to reach the rejection region and a real difference goes undetected.
- Power, $1 - \beta$, is the rest of the orange curve, the part past the critical values.

In Figure 4, μ_a is 2.5 standard errors above μ_0, which leaves $\beta = 0.29$ and power $= 0.71$. This design misses a real difference of that size in almost three tries out of ten. Slide the orange curve farther right, meaning a larger real difference, and the gray area shrinks. Move the red lines outward, meaning a smaller α, and the gray area grows.

Effect of sample size on errors

Sample size mainly affects Type II error. As n grows, the sampling distribution narrows, true effects are easier to detect, β falls, and power rises. With a fixed α and correct assumptions, the long-run Type I error rate stays at α for any n, because the critical values are placed so that the area past them under the H_0 curve is α at every sample size.

The actual Type I rate can differ from α in two cases. At small n, it can drift from α when the sampling distribution of the mean is not close to normal (Chapter 3), for example when the population is strongly skewed. Testing the data again each time new observations arrive also raises the Type I rate, at any n, because each new test is another chance for the sample mean to fall in the rejection region by chance.

Figure 5 shows the effect of n on Type II error with the same μ_0 and the same true μ. In both panels the sample mean is equally unlucky, 2.5 standard errors below μ. With the smaller sample, μ is 4 standard errors from μ_0, so $\bar{x}$ is 1.5 standard errors from μ_0, inside the critical value. The confidence interval still catches μ_0, and we fail to reject. With four times the sample, the standard error halves, because $\sqrt{4} = 2$. The same distance from μ_0 to μ is now 8 standard errors, so $\bar{x}$ is 5.5 standard errors from μ_0. The interval clears μ_0, and we reject H_0.

How to interpret *p*-values

A p-value is the probability, assuming H_0 is true, of obtaining a test statistic at least as extreme as the one we observed. We reject at level α when $p < \alpha$. α is chosen before the data are collected, and it sets how far out the critical values are. The p-value is computed from the data, and it measures how far out the observed test statistic is.

Tails. The p-value includes the area in the tail or tails that H_a points to. For a two-tailed test, add the areas in both tails. For a one-tailed test, use only the tail in the direction H_a states.

Warning: Three misreadings of a p-value

1. p is the probability of a test statistic at least this extreme if H_0 is true. It is *not* the probability that H_0 is true.
2. A tiny p is not "super significant." It means only that the data would be very unlikely if H_0 were true. The effect itself can still be tiny and not worth acting on, especially with a large n. Report an effect size and a CI so a reader can judge whether it matters.
3. A large p does not show that there is no effect. With small n, true effects can yield large p (low power).

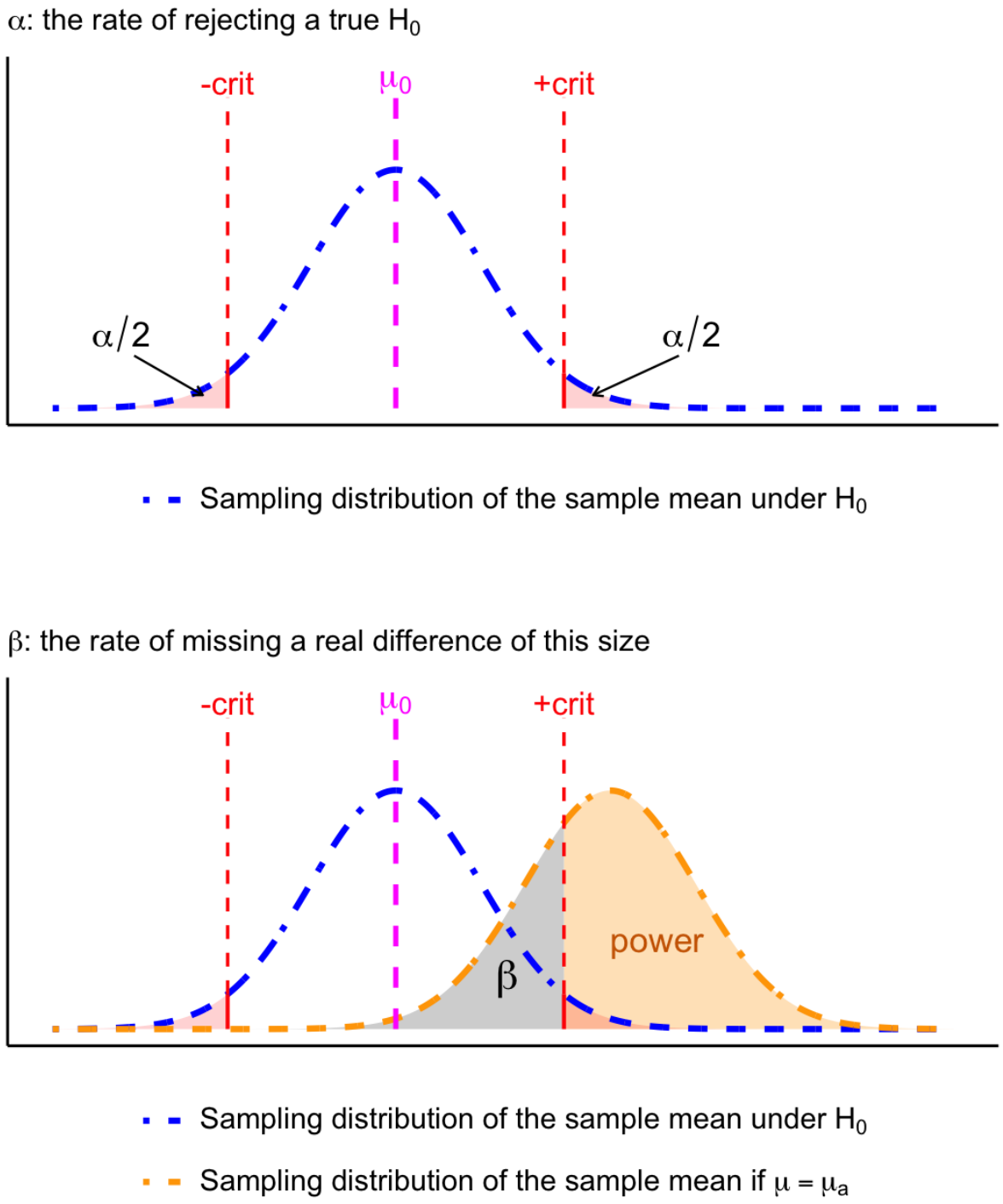


Figure 4: The two error rates, drawn on sampling distributions. Top: α is the red area in the tails of the curve under H_0. Bottom: the curve for one specific alternative value μ_a is added. β is the gray area under it between the critical values, and power is the orange area past the right critical value.

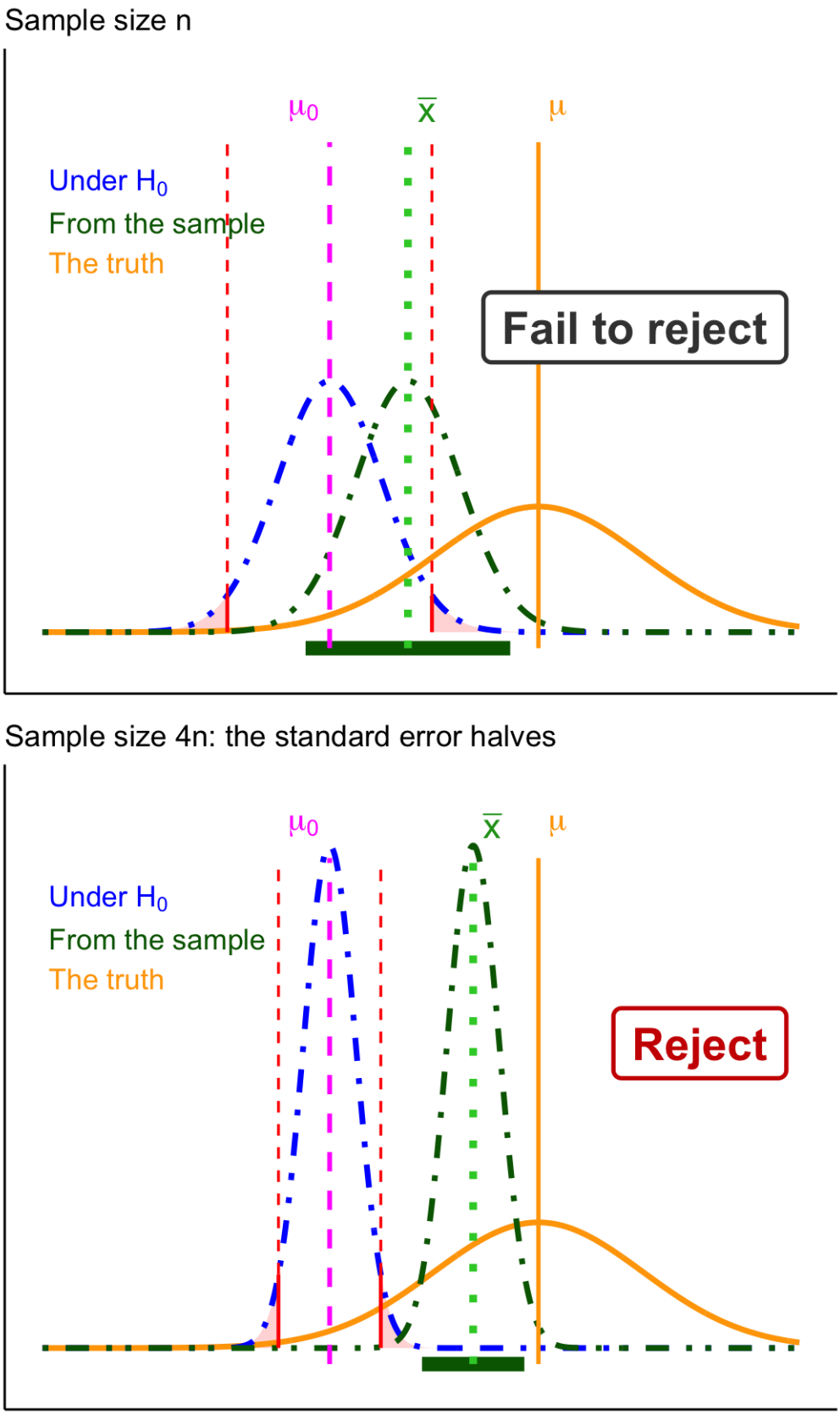


Figure 5: The Type II situation at two sample sizes, with the same μ_0, true μ and population. In both, $\bar{x}$ is 2.5 standard errors below μ. Top: the confidence interval catches μ_0, so we fail to reject. Bottom: with four times the sample, the standard error halves; the same bad luck leaves the interval clear of μ_0, and we reject.

Beyond the 0.05 cutoff

Effect size and power answer two questions that a decision at $\alpha = 0.05$ leaves open. People who act on a result, such as scientists, managers, and policymakers, need to know how large an effect is and how likely the study was to detect it.

Working definitions

- **Statistical significance:** a result is statistically significant when $p < \alpha$, which means the data would be unlikely if H_0 were true.
- **Effect:** the difference a study is about, such as how far apart two group means are, or how far a mean is from the value H_0 states (Chapter 4).
- **Effect size:** the number that says how large the effect is, in the study's units or in standard deviations.
- **Practical significance:** the effect is large enough to change what someone does about it. This is a judgment about the world, not a calculation. With a large enough n, a result can be statistically significant and practically trivial. With too small an n, a result can be practically important and still not statistically significant.
- **Power ($1 - \beta$):** the probability that your design correctly rejects H_0 when a real effect of a specified size exists, at your chosen α.

Choose the effect to detect and plan power before you sample. Decide what effect is *meaningful* for your context (e.g., a 0.3 mg/L nitrate reduction) and choose a sample size that gives adequate power, commonly 80%, to detect that effect.

One tail or two

The choice between a one-tailed and a two-tailed test changes power. Chapter 4 allowed a one-tailed test only when a difference in the other direction would not change your decision or action. Practical significance asks the same kind of question, whether a result would change what someone does. You can therefore answer Chapter 4's question before you collect data, by deciding which results would change your action. A one-tailed test is justified when

- the direction is set *before* seeing the data, and
- a result in the other direction would lead to the same action as no result.

A city testing whether a no-idling rule changed PM2.5 near schools would act on a rise just as much as on a drop, so that test stays two-tailed. For a plant checking whether its discharge exceeds a permit limit of 10 mg/L, a mean below the limit leads to the same action as a mean at the limit, which is to keep operating. That test can be one-tailed, with $H_a : \mu > 10$, as long as the plant wrote the direction down before sampling.

A one-tailed test puts all of α in one tail. Its critical value is the point with 5% of the H_0 curve beyond it, 1.645, where the two-tailed test uses the point with 2.5% beyond it, 1.96. Moving the critical value in from 1.96 to 1.645 makes a real effect in that direction easier to catch. For the design in Figure 4, β drops from 0.29 to 0.20 and power rises from 0.71 to 0.80. Figure 6 draws both tests on that design.

The cost is that a one-tailed test cannot detect an effect in the other direction at all, however large it is. In the bottom panel of Figure 6, the faint curve on the left is an effect of the same size in the other direction, and no rejection region lies under it.

Choosing the tail after looking at which way the data went amounts to a two-tailed test at $\alpha = 0.10$. If H_0 is true, the sample mean is equally likely to fall on either side of μ_0. Whichever side it falls on, you put a 5% rejection region there, so 5% of samples can be rejected in the right tail and another 5% in the left. The actual Type I error rate is then 0.10, double the 0.05 you report.

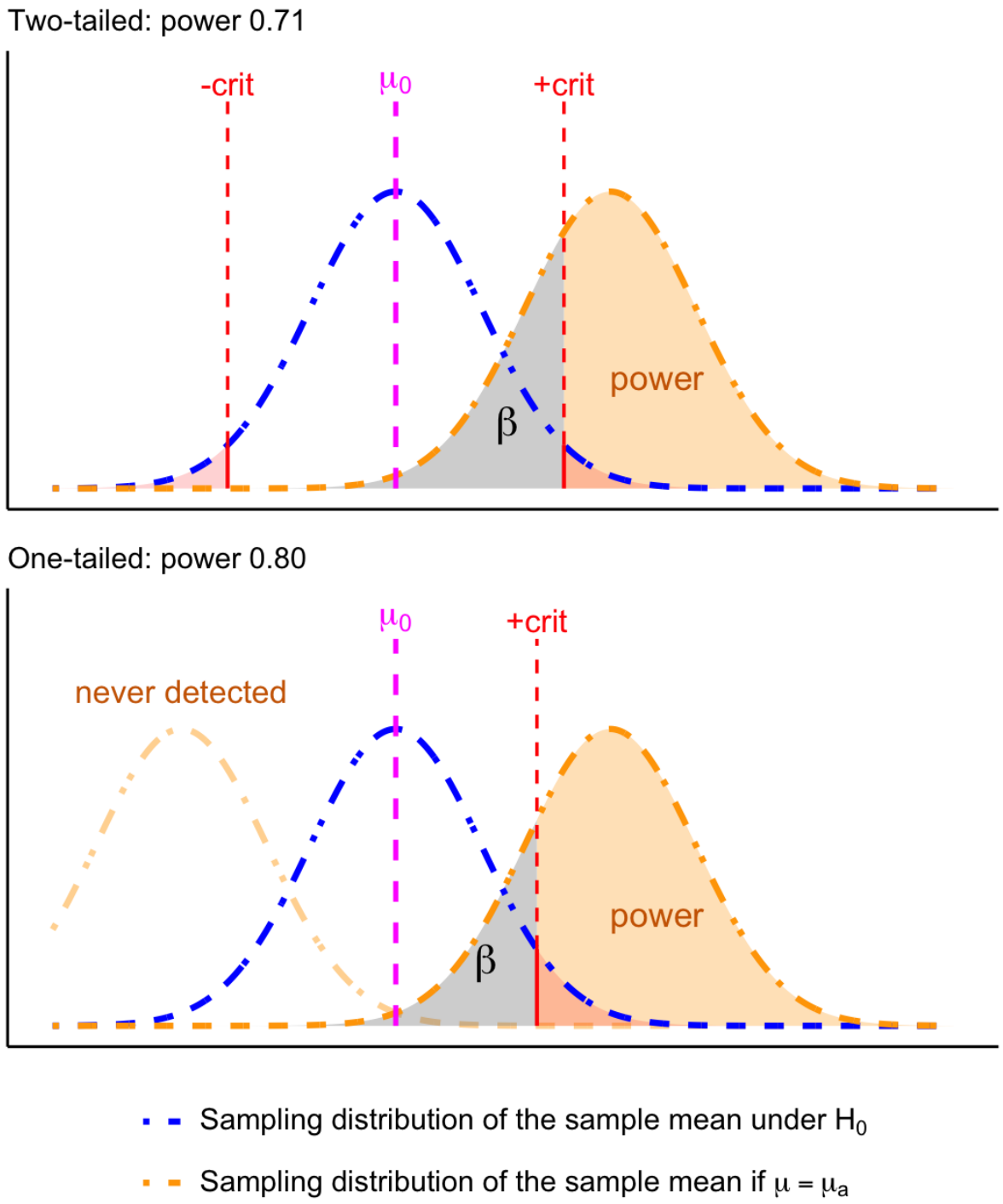


Figure 6: The same alternative under a two-tailed and a one-tailed test at $\alpha = 0.05$. Top: two-tailed, with $\alpha/2$ in each tail. Bottom: one-tailed, with all of α in the right tail, so the critical value moves in, the gray β area shrinks and the orange power area grows. The faint curve on the left is an effect of the same size in the other direction, which the one-tailed test never detects.

Effect size

Effect size comes in two kinds of units. You can express it in original units, which carry practical meaning, or in standardized units, which let you compare across scales.

Practical (original units). Use the scale stakeholders recognize.

- Exams: "Treatment improved scores by 5 percentage points" (half a letter grade).
- Time-on-task: "New training reduces practice time by 1,000 hours out of 10,000" (a 10% reduction).

These statements are easy to interpret, but they are hard to compare across contexts or measures. To say whether a 5-point exam gain is a larger effect than a 1,000-hour cut in practice time, you need each one measured against the variability of its own scale.

Standardized (unitless). When units are unfamiliar or scales differ, standardize by typical variability. The most common standardized effect size is **Cohen's** d (Cohen 1988). Cohen's d is the raw difference, Δ, divided by the population standard deviation, so it gives the effect in standard deviations:

$$\text{Cohen's } d = \frac{\Delta}{\sigma}$$

Δ is the effect in the study's units. It is $\mu_1 - \mu_2$ when comparing two groups, or $\mu - \mu_0$ when testing one mean against a reference value.

Cohen's d and z both put the effect in the numerator but divide by different things. Chapter 4 gave the general shape, effect over error, and filled it in with the z:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \qquad \text{versus} \qquad \text{Cohen's } d = \frac{\mu - \mu_0}{\sigma}$$

In z the numerator is the observed effect, $\bar{x} - \mu_0$. In Cohen's d it is the true effect, $\mu - \mu_0$, which $\bar{x} - \mu_0$ estimates. The test statistic divides by the standard error, which shrinks as n grows, so z gets larger with more data even when the effect stays exactly the same. Quadrupling n halves the standard error, so the same $\bar{x} - \mu_0$ gives twice the z. Cohen's d divides by the standard deviation, which does not shrink with n, so Cohen's d reports the size of the effect itself. Table 2 lines the two up.

Table 2: The test statistic z and the effect size Cohen's d compared.

	z	Cohen's d
Numerator	$\bar{x} - \mu_0$ (observed effect)	$\mu - \mu_0$ (true effect, estimated by $\bar{x} - \mu_0$)
Denominator	$\sigma/\sqrt{n}$	σ
As n grows	Grows for the same effect	Stays the same
Answers	Is the effect distinguishable from 0?	How large is the effect?

A large test statistic therefore does not mean a large effect. In a one-sample test, dividing by $\sigma/\sqrt{n}$ is the same as dividing by σ and multiplying by $\sqrt{n}$, so $z = \frac{\bar{x}-\mu_0}{\sigma}\sqrt{n}$. The fraction is Cohen's d computed from the sample, with $\bar{x}$ in place of μ, so $z = (\text{Cohen's } d)\sqrt{n}$. An observed Cohen's $d = 0.05$ gives $z = 0.05 \times \sqrt{100} = 0.5$ at $n = 100$ and $z = 0.05 \times \sqrt{10{,}000} = 5$ at $n = 10{,}000$. The t in Chapter 6 is the same ratio with an estimated s in place of a known σ, so the same holds for t.

Environmental example (biodiversity index). Suppose an index increases from 3 to 4 after reforestation.

- In raw units: +1 index point.
- In standardized terms:
  - If the SD across sites is 1.0, then Cohen's $d = 1.0$, a large shift by the usual guideline (0.8 or more is large, see the Tip below).
  - If the SD is 10, then Cohen's $d = 0.10$, a small shift (below the guideline's 0.2).

The same 1-point increase is a large effect where the index varies little across sites and a small one where it varies a lot.

Planning starts from Δ. To plan a study, choose the value of Δ to plan for, in the units of the study. It might be the effect you expect to see, or the smallest difference that would change what you do. A power analysis also needs an estimate of σ, and with Δ and σ you have Cohen's $d = \Delta/\sigma$.

> Tip: Estimating and reporting Cohen's d in practice
>
> - With measured data, σ is unknown, so estimate it from the sample. For two groups, use the pooled SD from Chapter 6.
> - Heuristics (context matters): Cohen's $d \approx 0.2$ (small), 0.5 (medium), 0.8 (large).
> - A "small" Cohen's d can still be important (e.g., small reductions in lead or PM2.5 can matter for health).
> - Report the effect in the study's units with a 95% CI, to show how precisely it was estimated. Report Cohen's d as a single number.

Suppose we're measuring final exam performance (percent correct). The class mean is 65% with a standard deviation of 5 percentage points. Group A is a control, while Group B receives some instructional treatment.

Figure 7 shows four possible scenarios, with Group B's mean shifted by Cohen's $d = 0$, 0.5, 1 and 2.

In Figure 7, a half-SD shift (Cohen's $d = 0.5$, 2.5 points) leaves the two curves mostly overlapping, and a two-SD shift (Cohen's $d = 2$, 10 points) separates them.

In practice, we rarely know beforehand how big a shift to expect, so the power curves later in this chapter show power across a range of effect sizes.

Power

A design's power is the probability that it correctly rejects H_0 when a real effect of a specified size exists, at your chosen α. Effects vary in size, and designs vary in power. A design with low power will usually fail to reject H_0 even when a real effect exists.

Power rises when:

1. Effect size is larger.
2. Sample size (n) is larger (smaller standard error).
3. The cutoff for rejecting H_0 is less strict (larger α).
4. Variability in the population is lower (smaller σ), for example from less measurement noise.

When you plan a study, you need to know how large n must be to reach 80% power, with α fixed at 0.05, for the smallest effect that matters. Figure 10 answers that question for one effect size.

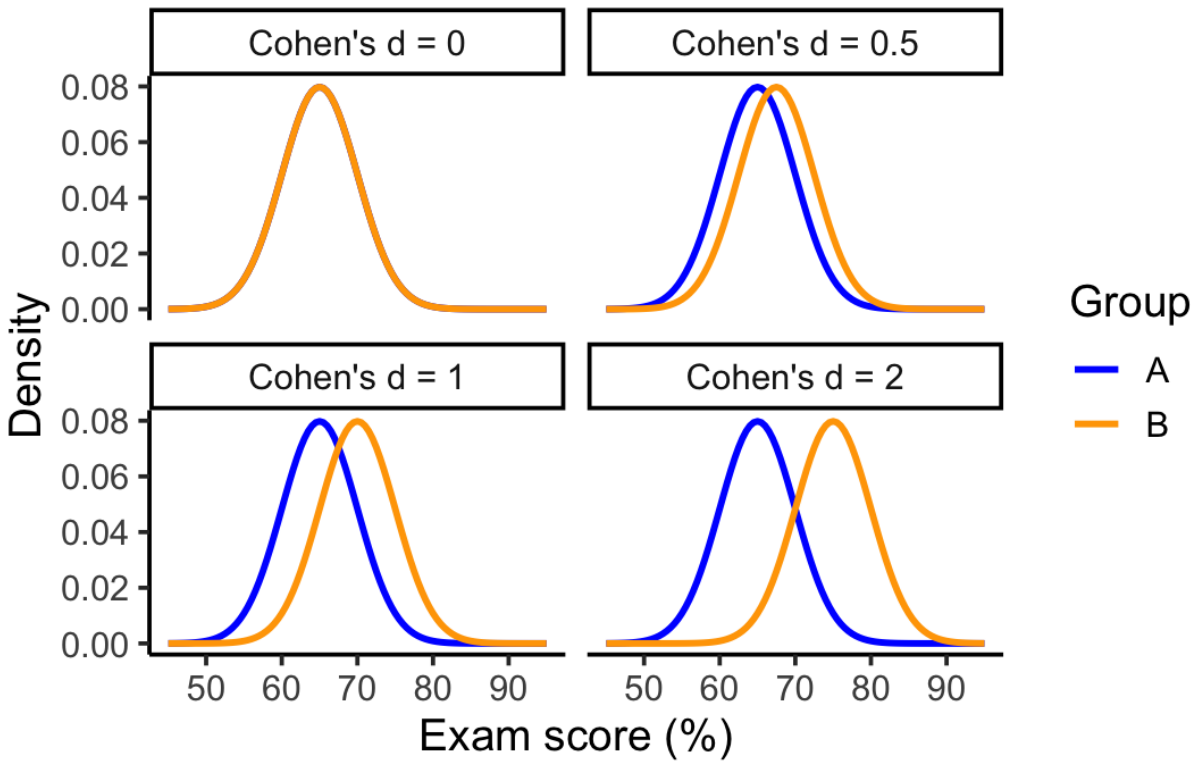


Figure 7: Exam scores for a control group A (mean 65, $\sigma = 5$) and a treatment group B whose mean is shifted by Cohen's $d = 0$, 0.5, 1 and 2, that is, by 0, 2.5, 5 and 10 points.

Estimating power by simulation

A simulation with the exam-score example shows how design choices change power. We simulate two groups of $n = 10$ each. We draw Group A (control) from a normal distribution with mean 65 and $\sigma = 5$, and Group B (treatment) from one with mean 67.5, a shift of 2.5 points, or Cohen's $d = 0.5$, a medium effect. These are the two distributions in Panel 2 of Figure 7.

Each simulated experiment compares the two groups with an independent-samples t-test. Chapter 6 covers this test. For now, treat it as a test that returns a p-value for the difference between two group means, the same job the z statistic in Chapter 4 does for one mean.

Before you look. One simulated experiment draws 10 scores for Group A and 10 for Group B, runs the t-test, and saves the p-value. The simulation repeats this 1,000 times and counts how often $p < 0.05$, which is how often the test rejects H_0 at $\alpha = 0.05$. In every one of the 1,000 experiments, Group B's true mean is 2.5 points higher. Decide roughly how often you expect the test to detect that difference with 10 per group.

The test detected the 2.5-point difference in about 22% of the 1,000 experiments. The proportion of simulated experiments that reject H_0 estimates the design's power, the probability that the test detects a real difference of this size. With $n = 10$ per group, the test misses this medium-sized effect in about 78% of experiments.

How power changes

Each of the next three simulations changes one part of the previous design and reruns the 1,000 experiments.

1. **Larger n.** Doubling to $n = 20$ per group divides the standard error of each group mean by $\sqrt{2}$, about 1.41, because the standard error is $\sigma/\sqrt{n}$ and $\sqrt{20}/\sqrt{10} = \sqrt{2}$.

That increases power to about 35%, up from about 22% at $n = 10$ (rows 1 and 2 of Table 3).

2. **Smaller α.** Making the cutoff stricter (e.g., $\alpha = 0.01$) lowers power, because only experiments with $p < 0.01$ now count as significant. We reuse the same $n = 20$ simulations and change only the cutoff.

Power falls to about 14% (row 3).

3. **Larger effect.** If the treatment produces a 2-SD shift, Group B's mean is $65 + 2 \times 5 = 75$. We rerun the simulation with the same design at the same $\alpha = 0.01$.

The test rejected H_0 in 100% of the 1,000 experiments (row 4). With a 2-SD difference, rejecting H_0 is close to certain, even with 20 per group and $\alpha = 0.01$.

Table 3 collects the four designs. Each row gives the simulated power and, next to it, the exact power for the same design, computed from the t distribution.

Table 3: Simulated and exact power for the four designs. Group A's mean is 65 and $\sigma = 5$ in every design.

n per group	α	Group B mean (Cohen's d)	Simulated power (1,000 runs)	Exact power
10	0.05	67.5 (0.5)	22%	19%
20	0.05	67.5 (0.5)	35%	34%
20	0.01	67.5 (0.5)	14%	14%
20	0.01	75 (2)	100%	99.97%

Table 4 shows which way each change moves α, β and power, and Figure 8 draws each change on the design from Figure 4.

Table 4: How each change moves α, β and power.

Change	α	β	Power	In Figure 8
Larger n	Same	Down	Up	(a)
Smaller α	Down	Up	Down	(b)
Larger α	Up	Down	Up	(c)
Larger effect	Same	Down	Up	(d)

Power curves

A design's power depends on how large the true effect is. A **power curve** plots the power of a design against one quantity that varies, such as the true effect size or the sample size, with everything else held fixed. Figure 9 fixes the design at an independent-samples t-test with $n = 10$ per group and $\alpha = 0.05$, and plots power against the true effect size, Cohen's d.

With 10 per group, this design detects a medium effect (Cohen's $d = 0.5$) only about 19% of the time, a larger effect (Cohen's $d = 0.8$) about 40% of the time, and a very large effect (Cohen's $d = 2$) almost every time (about 99%). Only effects of about Cohen's $d = 1.3$ or larger reach 80% power with this design.

The simulation above estimated power at Cohen's $d = 0.5$ as about 22%. The two numbers differ because 1,000 simulated experiments give an estimate, while the curve is computed exactly.

The second curve holds the effect size fixed and varies the sample size. Suppose prior research suggests the effect is small (Cohen's $d = 0.2$). Figure 10 plots power against sample size for Cohen's $d = 0.2$, with a dashed line at 80% power, a common planning target.

At 10 per group, power to detect Cohen's $d = 0.2$ is about 7%. Reaching 80% power takes about 394 per group. At that sample size, 20% of studies would still fail to reject H_0 for a real effect of Cohen's $d = 0.2$.

In a power analysis (defined in the next section), the n that reaches a power target such as 0.80 depends on the effect size and standard deviation you assumed. The 394 per group above assumes Cohen's $d = 0.2$, and assuming Cohen's $d = 0.3$ instead gives 176 per group. Report the assumed effect size and standard deviation alongside the n.

Planning your design

Planning a study means working through the following, in order:

- Choose the Δ to plan for, the smallest difference that would change what you do, stated in the units of the measurement.
- Choose α (often 0.05; justify if different).
- Estimate variability, σ (pilot data, literature, or historical monitoring). With Δ and σ you have Cohen's d, which is what a power analysis needs.
- Choose one- or two-tailed before data collection. The choice changes the critical value, so it changes the n you need.
- Compute n for $\geq 80\%$ power (or explain tradeoffs if you can't reach it).

This sequence is a **formal power analysis**. A formal power analysis finds the sample size n that gives a target power, usually 80%, for a chosen Δ, σ, α and number of tails, before any data are collected. It applies while you can still choose n, for example when deciding how many plots to establish or how many samples a permit application commits you to. When the data already exist, as in your course project, n is fixed. Report the effect and its confidence interval instead.

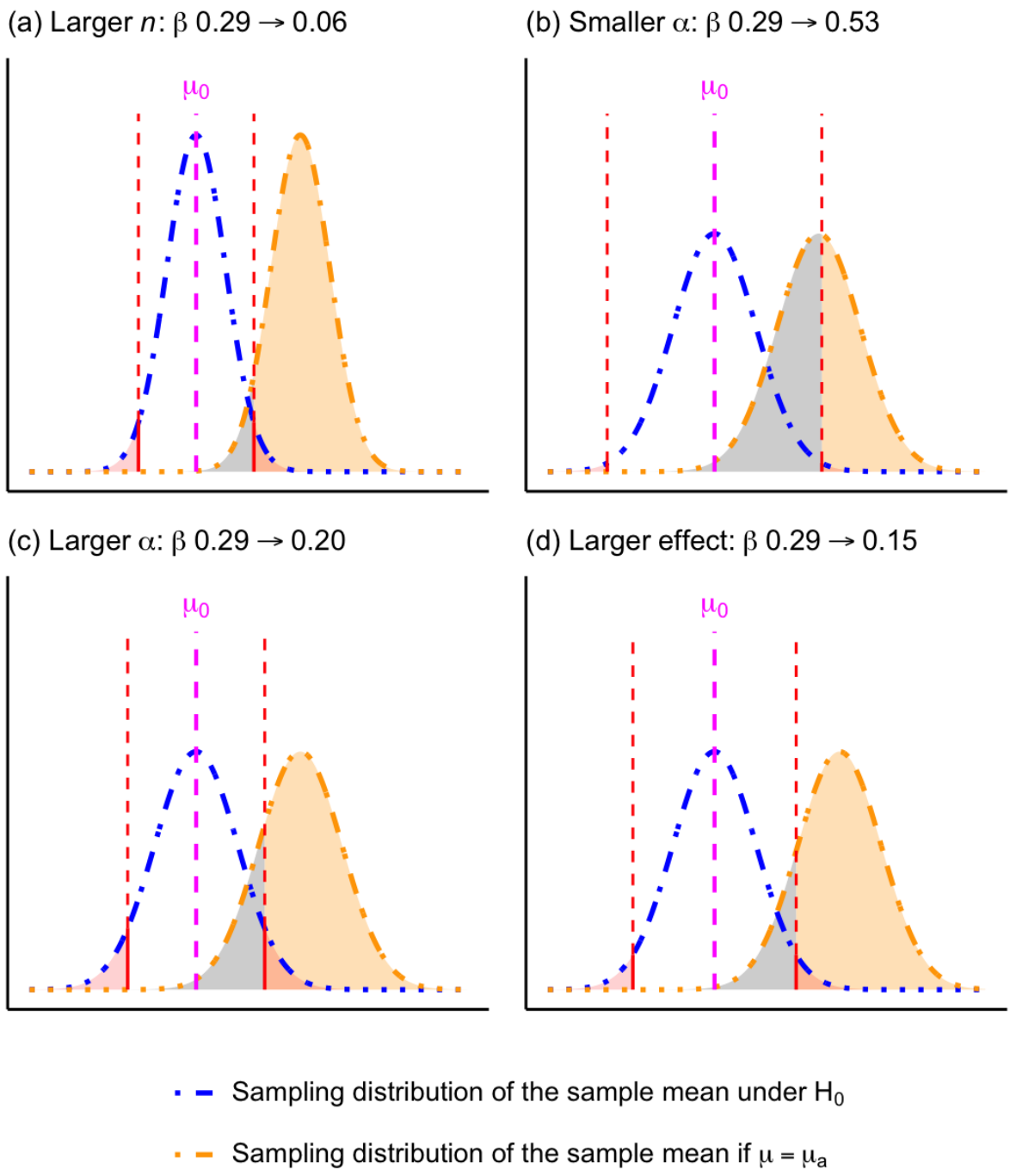


Figure 8: Each change from Table 4, drawn on the design in Figure 4 (μ_a 2.5 standard errors above μ_0, $\alpha = 0.05$). Red is α, gray is β, and orange is power, as in Figure 4. Each panel title gives β for the original design and after the change. (a) Twice the sample size narrows both curves. (b) A smaller α moves the critical values out. (c) A larger α moves them in. (d) A larger effect moves the orange curve farther from μ_0.

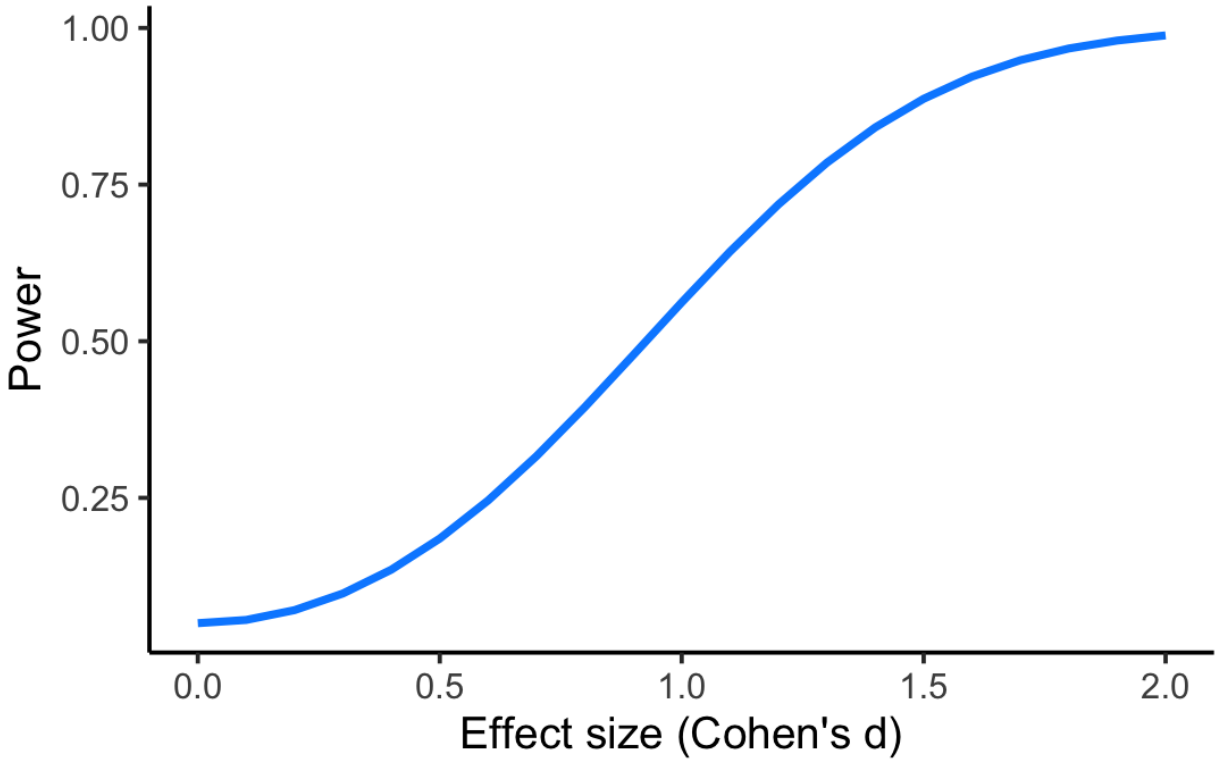


Figure 9: Power vs. effect size (Cohen's d) for an independent-samples t-test with $n = 10$ per group and $\alpha = 0.05$.

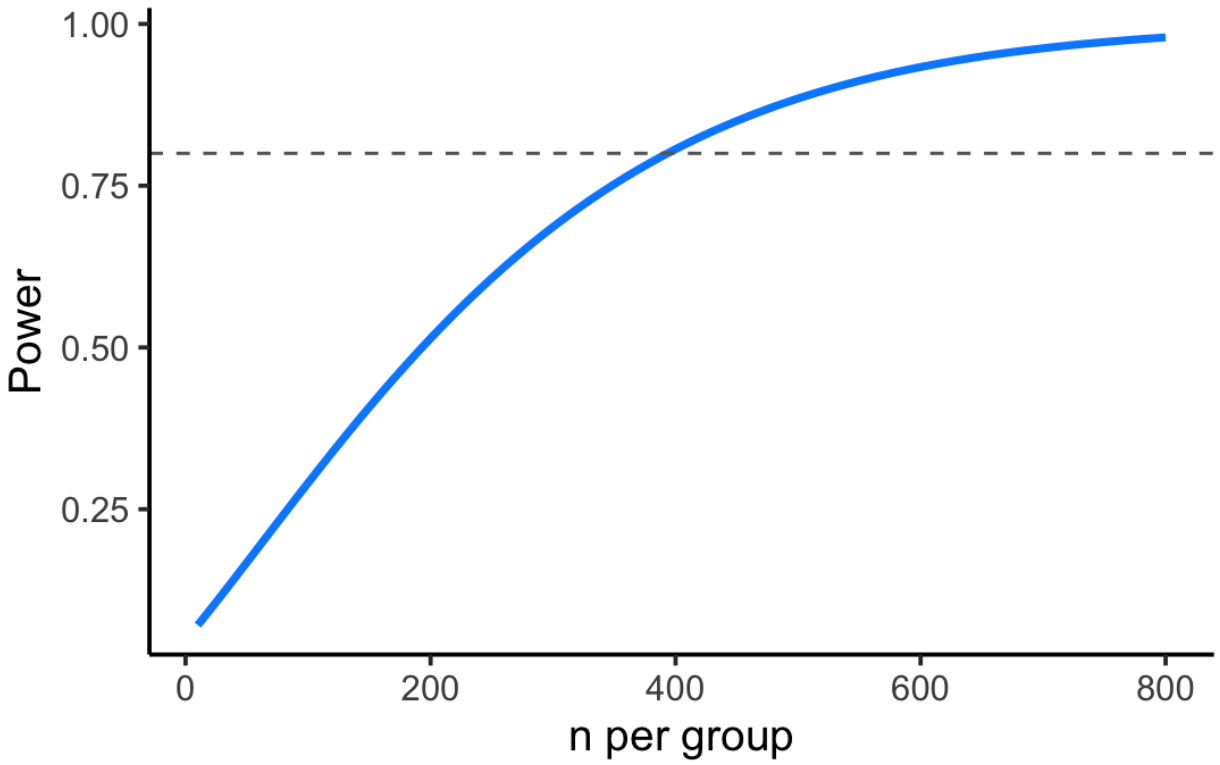


Figure 10: Power vs. sample size for detecting Cohen's $d = 0.2$ with an independent-samples t-test at $\alpha = 0.05$. The dashed line marks 0.80 power.

Warning: Post-hoc power

Post-hoc power is power computed after a study, using the effect size the study observed. It is a restatement of the p-value, so a non-significant result always has low post-hoc power. For a two-tailed z-test, a result with $p = 0.05$ exactly has post-hoc power of about 50%, and any larger p gives less. Do not compute it. Power for a difference chosen in advance, such as the smallest effect that matters, is fine to report.

Low power and very large samples

- **A significant result from a low-power study often fails to replicate.** A design with 30% power to detect an effect that is really there rejects H_0 in only about 3 of 10 repeats, so a significant first result is usually followed by non-significant replications.
- **A significant result from a small study overstates the effect.** When a small study does reach significance, its estimated effect size tends to be too large. When the null is true, that large estimate makes a false positive look convincing.
- **Huge n finds tiny effects.** With very large samples, trivial differences can register as statistically significant. For example, a utility comparing two billing designs across 2 million households can find a 0.1% difference in electricity use statistically significant, even though a difference that small would not change which design it adopts. Report an effect size and a confidence interval alongside each p-value so a reader can judge whether a difference matters.

Figure 11 shows that a significant result from a small study overstates the effect. Each dot in the figure is one simulated study in which the true mean is exactly μ_0. We ran 1,000 studies at each of four sample sizes (10, 50, 100, and 1,000) and kept only the ones that gave $p < 0.05$, about 50 at each size.

When the null hypothesis is true, 5% of tests still give $p < 0.05$. In small samples, those false positives come with large estimated effect sizes, even though the true effect is zero. A one-sample t-test (Chapter 6) at $n = 10$ reaches $p < 0.05$ only when $|t| > 2.26$, the t critical value for this sample size. Because $t = (\text{Cohen's } d)\sqrt{n}$ as well (the effect size section), that means $|\text{Cohen's } d| = |t|/\sqrt{10}$ must be at least 0.72. At $n = 1{,}000$ the threshold is 0.06.

Reporting results

A results statement is one sentence that states the finding in plain language and follows it with the supporting statistics, separated by commas.

Template: *Plain-language finding*, statistic with degrees of freedom (df, Chapter 6) or n, p-value, effect size and CI.

What to include (in order):

1. The finding in plain language, with its direction.
2. The test name, when the statistic does not already identify the test (for example, a rank test or Welch's t).
3. Test statistic and its degrees of freedom (or sample size for z).
4. Exact p.
5. The effect in the study's units with its 95% CI (Chapter 6 shows how to build it for a mean difference), and Cohen's d where the chapter that covers the test reports it.

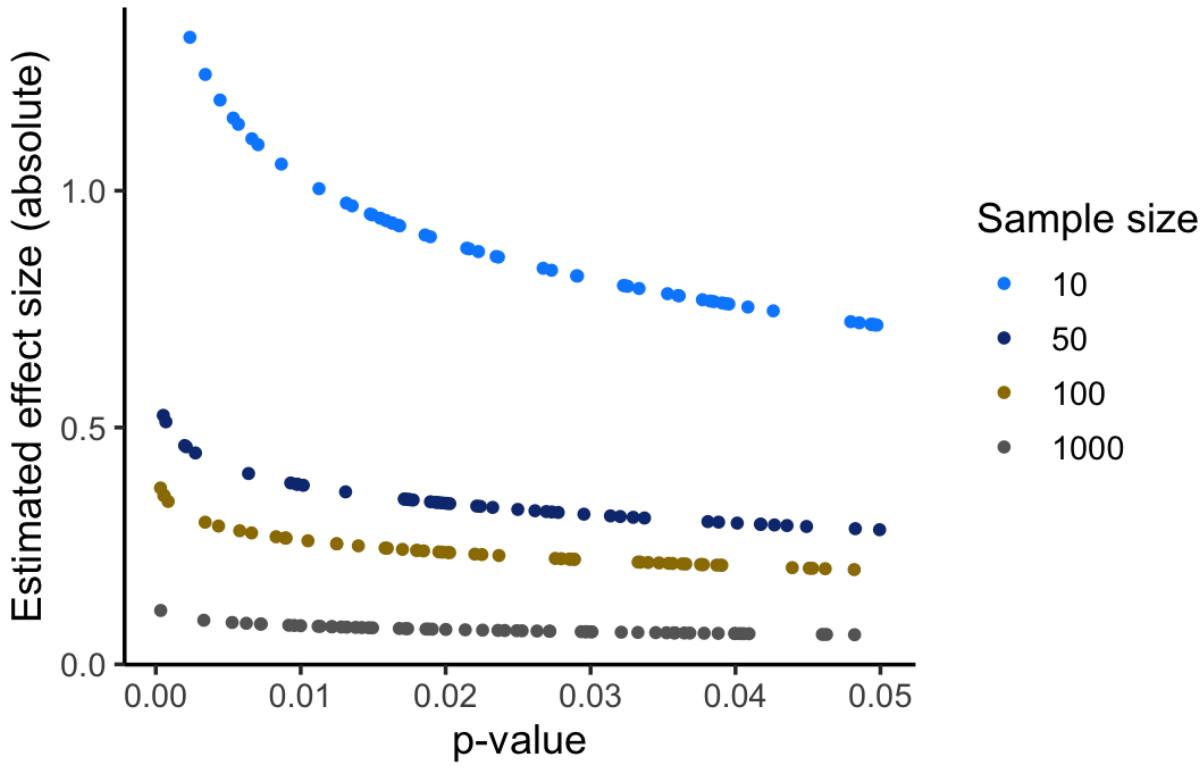


Figure 11: Estimated effect size among Type I errors ($p < 0.05$) under the null, by sample size, for a one-sample t-test. Effect sizes are shown as absolute values, since a false positive is equally likely in either direction and the sign carries no information here. Small n inflates estimated effects.

Examples

The t-tests in these examples are covered in Chapter 6.

- **One-sample z.** The average mercury level within 10 km of the smelter, 0.62 mg/kg, was 0.12 mg/kg higher than the legal limit of 0.50 mg/kg, $z = 2.85$, $n = 32$, $p = 0.004$, 95% CI [0.04, 0.20] mg/kg.
- **One-sample t with CI.** Mean nitrate exceeded the target by 0.41 mg/L, $t(31) = 2.60$, $p = 0.014$, 95% CI [0.09, 0.73] mg/L.
- **Independent-samples t with effect size.** Mean turbidity was 4.0 NTU higher at downstream sites than at upstream sites, $t(58) = 3.10$, $p = 0.003$, 95% CI [1.42, 6.58] NTU, Cohen's $d = 0.80$.
- **Not significant (state "fail to reject").** Mean canopy temperature did not differ significantly from the baseline, mean difference 0.30 °C, $t(47) = 1.20$, $p = 0.236$, 95% CI [-0.20, 0.80] °C, so we fail to reject H_0.

Rounding and style

- Round test statistics to 2 decimals and p to 3 decimals (use $p < 0.001$ when needed).
- Use "fail to reject H_0" (not "accept").
- State the direction the data went ("higher than," "lower than") when the test rejects, and "did not differ from" when it fails to reject.

Do / Don't

- Do follow the sentence with one sentence on what the result means for the question.
- Do give units for every value.
- Do not restate methods.
- Do not claim proof.

Chapter Summary

Why it matters. A hypothesis test ends in reject or fail to reject, and that decision can be wrong in two ways. Effect size tells you how large the difference is, and power tells you whether the study could have detected it. Chapter 6 applies all of this to the t-tests.

Core ideas

- **Two ways to be wrong.** A Type I error rejects a true H_0, at a rate you set with α. A Type II error fails to reject a false one. For a fixed design, lowering α trades fewer Type I errors for more Type II errors. $\alpha = 0.05$ is a widely used convention for where to make that trade.
- **What a p-value measures.** It is the probability, if H_0 were true, of a test statistic at least as extreme as the one we observed. It is not the probability that H_0 is true, and it says nothing about how large an effect is.
- **Significance and effect size answer different questions.** A significant result says the data would be unlikely if H_0 were true. The standard error shrinks as n grows, so the same small difference produces a larger and larger test statistic. With a large enough n, even a trivially small difference is likely to be significant. Report an effect size and a confidence interval next to every p-value so a reader can judge whether the result matters.
- **One tail only when the other direction would not change the action.** Choose a one-tailed test before seeing the data, and only when a result in the other direction would lead to the same action as no result. Choosing the tail afterward doubles the actual Type I error rate.

- **Effect size in two kinds of units.** The same difference can be written in original units, Δ, which carry practical meaning for the people who act on your result, or in standard deviations, Cohen's $d = \Delta/\sigma$, which makes results comparable across studies and scales. Report both when you can, and plan in Δ, because the smallest difference worth detecting is a judgment made in the units of the measurement.
- **Power is set by the design.** Power is the probability that a design correctly rejects H_0 when a real effect of a specified size exists. It depends on n, effect size, α, and variability together. An underpowered study usually misses real effects, and when it does reach significance, the effect it reports tends to be too large. Run a power analysis before collecting data, while n can still be chosen.
- **Report a result as a sentence.** State the direction in words, then give the statistic with its df or n, the p-value, and the effect in the study's units with its 95% CI, all separated by commas. Add Cohen's d where the chapter that covers the test reports it, and name the test only when the statistic does not identify it.

Cohen, J. 1988. *Statistical Power Analysis for the Behavioral Sciences*. Second. Lawrence Erlbaum.